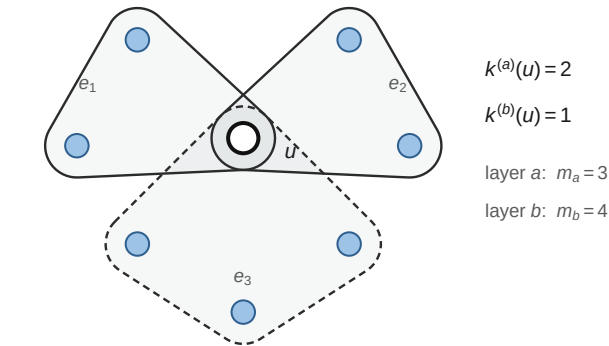


(a) Multiplex hypergraph and hyperdegrees

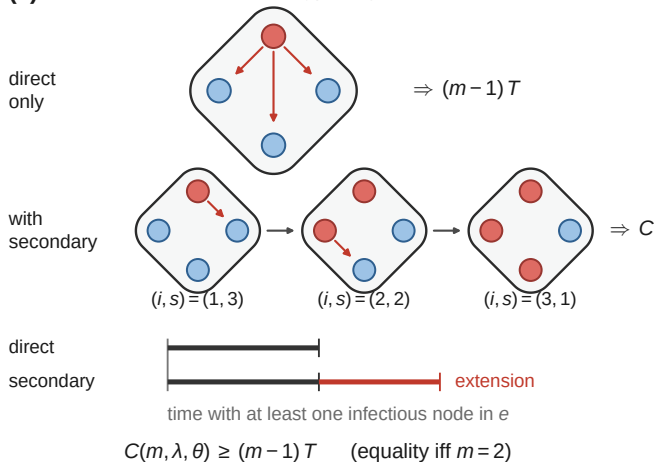


joint hyperdegree distribution $P(\mathbf{k})$, $\mathbf{k} = (k^{(1)}, \dots, k^{(M)})$

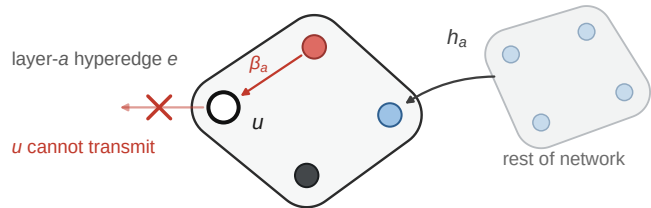
$\Psi(\mathbf{x})$ — uniformly sampled node [Eq. (3)]

$\psi_a(\mathbf{x})$ — node reached via layer a , arrival edge excluded [Eq. (4)]

(c) Rare-infection limit I: within-hyperedge factor C_b

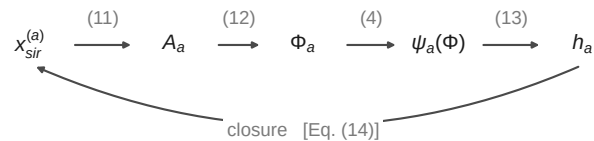


(b) Edge-based closure at finite prevalence



$x_{sir}^{(a)}$ — joint SIR composition of the m_a-1 other members [Eq. (9)]

$\Phi_a - e$ has not infected u ; $A_a - e$ is transmitting [Eqs. (10)–(12)]



$\Rightarrow S(t), I(t), R(t)$ and $R(\infty)$ [Eqs. (8), (17), (18)]

(d) Rare-infection limit II: excess hyperdegree B_{ab}

